



Nonlinear modeling, Measurement, and Control of a hybrid reluctance-actuated fast steering mirror

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ABSTRACT

The hybrid reluctance-actuated fast steering mirror (HRA-FSM) addresses the performance limitations of conventional fast steering mirrors in terms of aperture, bandwidth, and stroke. However, the nonlinear torque characteristics that arise at large deflection angles continue to constrain further improvements in HRA-FSM performance. Existing mathematical models exhibit considerable errors, and current studies lack effective methods for nonlinearities evaluation. In this study, a novel nonlinear HRA model is developed based on a prototype system. This model establishes the mapping between nonlinear common-mode and differential-mode components to elucidate the complex angle-current coupling dynamics. Experimental validation is performed by quantifying magnetic flux density distribution differences at the stator end faces and Hall sensor measurements. This paper details the theoretical framework for nonlinear modeling, the experimental measurement principles, and introduces a feedback linearization compensation control algorithm. This approach achieves a 48.8% reduction in bandwidth fluctuation and up to 36.4% decrease in tracking error. The HRA-FSM prototype demonstrates a range-bandwidth product of 18,675 mrad Hz, surpassing the performance of existing single-axis FSMs.

1. Introduction

In the field of aviation optical remote sensing, the demand for high-dynamic imaging is increasingly prominent [1]. The fast steering mirror (FSM) serves as a core precision component, fundamentally determining both optical axis stabilization accuracy and imaging quality. However, conventional FSM actuation technologies face inherent limitations in dynamic performance. Specifically, the acceleration capacity of the voice coil motor (VCM) is inherently limited by its relatively low Lorentz force [2], making it inadequate for rapid-response applications under high-dynamic, heavy-load conditions. The piezoelectric actuator (PEA) is restricted in deflection angle due to the intrinsic characteristic of deformation ($\approx 0.1\%$) [3]. Although flexible displacement amplification mechanisms can improve stroke, they inevitably compromise system bandwidth [4,5]. Furthermore, integrating large-aperture mirrors (>50 mm) presents additional challenges to achieving both wide deflection angles and high bandwidth [6] in FSMs.

The hybrid reluctance actuator (HRA), which leverages the Maxwell stress principle, improves acceleration by 1–2 orders of magnitude over

VCMs [7] while maintaining millimeter-level stroke compared to PEAs. As a result, the HRA achieves an optimal trade-off between range and bandwidth [3], showing increasing potential in precision positioning stages [8–10], adaptive optics [11,12], vibration dampers [13,14], atomic force microscopy [15–17] and fast tool servos [18–20]. In HRA-FSM applications, [21] pioneered an HRA-based Advanced Fast Steering Mirror (AFSM) with 10^5 rad/s² acceleration and a 10 kHz small-signal bandwidth but limited range (± 3.5 mrad) and beam-angle constraints. [22] enhanced the design with an integrated magnetic circuit, achieving a ± 52 mrad range and a 1 kHz bandwidth. Later iteration [23] achieved a 1.5 kHz bandwidth and an acceleration of 5.3×10^4 rad/s², though eddy current sensor's resolution limited precision to 34 μ rad RMS. [24,25] demonstrated an HRA-FSM with ± 35 mrad range, 1.7 kHz bandwidth, and optical jitter below 1.5 μ rad RMS. [26] presented a novel magnetic circuit design but lacked experimental validation. [27] integrated strain sensors into HRA-FSM, enabling ± 1.5 mrad deflection in a compact layout. Despite these advancements, most HRA-FSMs have optical apertures smaller than

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25 mm, according to Rayleigh criterion ($\theta = 1.22 \frac{\lambda}{D}$), this fundamentally limits the resolution of optical systems.

A significant challenge for HRAs is their intrinsic nonlinear torque characteristics. While most studies adopt linearized torque models [21, 22, 26, 28], these introduce substantial errors at large angular strokes, thereby limiting control performance. Some studies have proposed advanced control algorithms [29, 30], while several studies have investigated HRA nonlinearities. For translational HRAs, polynomial functions have been employed to characterize nonlinear negative stiffness [16] and approximate leakage flux coefficients [31], while nonlinear negative stiffness and motor constants have also been derived from permanent magnet (PM) flux analysis [10, 32]. For deflection HRAs, nonlinearities have been modeled using polynomial expressions for the PM flux leakage coefficient [28, 33]. However, these polynomial-based methods require complex parameterization and exhibit poor generalizability. Moreover, the above studies neglect additional nonlinearities of magnetic flux under coil excitation.

Currently, there is no reliable methodology for evaluating output torque characteristics of deflection HRAs. This deficiency stems from two primary factors. First, traditional torque measurement methods assume rigid-body kinematics with a fixed rotation axis are incompatible with compliant mechanisms, where deflection arises from elastic deformation and instantaneous rotation centers vary dynamically. Second, traditional FSM deflection control utilizes discrete actuation units whose dynamics can be evaluated independently, whereas HRA-FSMs employ integrated magnetic circuit designs and compact configurations that complicate torque assessment. As a result, direct measurement introduces significant computational and installation errors.

To address the performance limitations caused by inaccuracies in linear torque models within HRA-FSM systems, this study comprehensively investigates the nonlinear torque characteristics of HRAs. Key contributions include:

1. A novel nonlinear torque model is developed by analyzing the nonlinearities of both permanent magnet flux and coil flux – factors neglected in linear models – and by establishing the mapping between common-mode and differential-mode flux components.
2. An innovative torque measurement methodology using linear Hall sensors is proposed. Optimal measurement regions are identified through analysis of stator-end magnetic flux density disparities, and the methodology is successfully validated experimentally.
3. A torque feedback linearization control algorithm is implemented, compensating the nonlinear system to achieve equivalence to a second-order linear system. This approach mitigates variations in HRA-FSM control performance across different deflection angles.

This paper is organized as follows: Section 2 introduces the proposed HRA-FSM. Section 3 describes the PM-biased analytical model. Section 4 identifies coil flux nonlinearities to formulate new nonlinear model. The mechanical design is described in Section 5. Section 6 details FEA-guided Hall sensor measurements for experimental validation. Section 7 presents the feedback linearization controller and its experimental verification. Conclusions are provided in Section 8.

2. System description

2.1. Composition and working principle

Fig. 1(a) shows a cross-sectional view of the single-axis HRA-FSM developed in this study. The FSM features an elliptical surface measuring 50.8 mm × 71.8 mm to accommodate both a 50 mm clear aperture and 45° beam incidence requirements. The actuator is mounted at the bottom, while the mover forms a deflection component with mirror

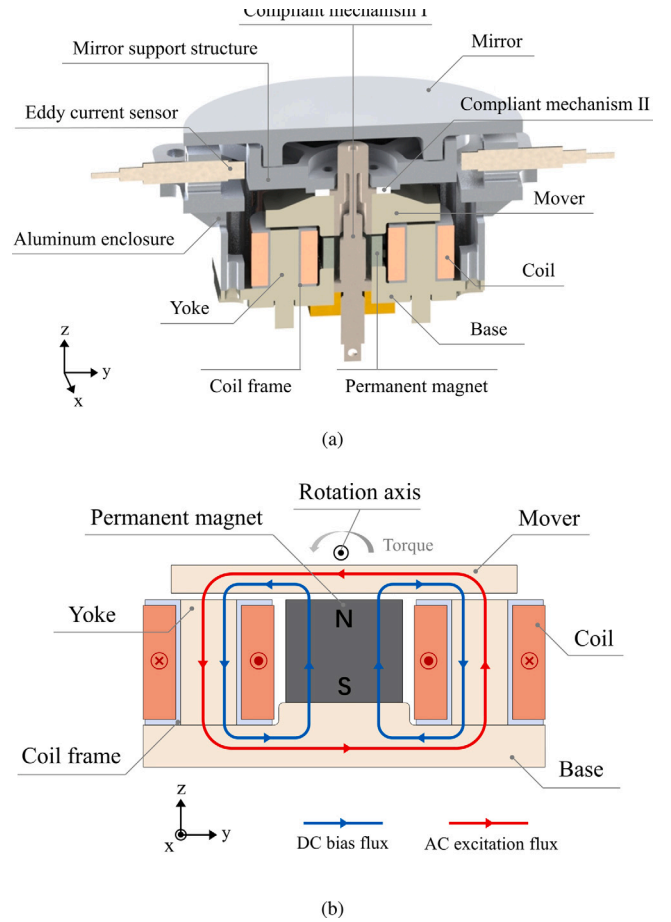


Fig. 1. (a) Cross-sectional view of the single-axis HRA-FSM. (b) Driving principle diagram of HRA.

and support structure. The compliant mechanism consists of two parts (for detailed discussion, see Section 5), allows frictionless deflection, compensates for negative stiffness, and restricts non-driven degrees of freedom. The sensing system employs a differential symmetrical layout with externally integrated eddy current sensors, ensuring precise position detection and efficient use of internal space.

Fig. 1(b) illustrates the actuator's working principle. The actuator consists of a magnetically conductive mover-yoke-base assembly forming a closed magnetic circuit. A centrally placed PM generates DC bias flux (blue lines), while the coil produces AC flux (red line) when excited by current. The superposition of these fluxes alters the magnetic field strength in the working air gap. According to the Maxwell stress tensor, this imbalance causes asymmetric stress at the mover's ends, resulting in mirror deflection.

2.2. Actuator design

Magnetic flux leakage, which is intrinsically related to the actuator's structural characteristics, significantly influences performance. Finite element analysis (FEA) was performed using Maxwell to evaluate the theoretical model, as shown in Fig. 2(a). The soft magnetic material employed was 1J22 iron-cobalt alloy, whose high saturation flux density (2.35 T) enables high output torque. The permanent magnet was made of NdFeB (grade N38SH, $B_r = 1.25$ T at 20 °C), and the magnetic pole area was designed to be slightly larger than the combined cross-sectional area of the yokes to maintain a high bias flux density and enhance energy utilization efficiency. The coil, with a diameter of 0.4 mm, consisted of 230 turns and a filling factor of 0.69. With

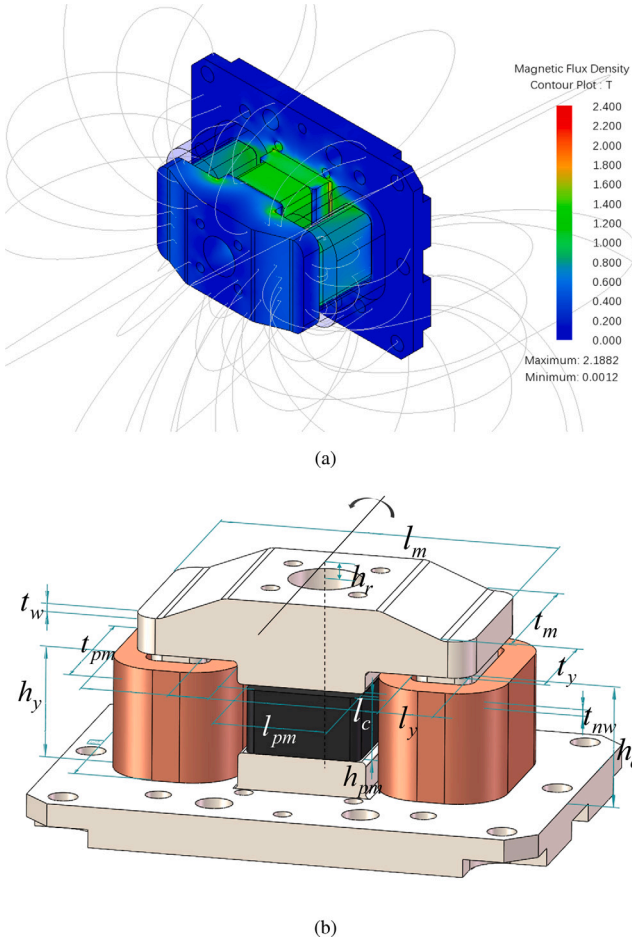


Fig. 2. (a) The geometric structure of the actuator. (b) Magnetic field distribution and leakage (gray line) of actuator.

Table 1
Dimension of the actuator.

Parameter	Value	Parameter	Value
Mover length (l_m)	37 mm	Width of yoke (t_y)	14 mm
Mover width (t_m)	20 mm	Coil thickness (l_c)	3.5 mm
PM width (l_{pm})	19 mm	Coil height (h_c)	12.2 mm
PM height (h_{pm})	7 mm	PM air gap (t_{nw})	0.75 mm
PM length (l_{pm})	12.5 mm	Working air gap (t_w)	0.95 mm
Yoke length (l_y)	6 mm	Rotation height (h_r)	1.05 mm
Yoke height (h_y)	13.3 mm	Torque length (h)	14.5 mm
Pole area (S_m)	186.4 mm ²	Yoke face area (S_y)	82.1 mm ²

reference to Fig. 2(b), the structural parameter table of the actuator is presented in Table 1.

3. Nonlinear analysis under PM bias condition

The conventional linear torque model assumes a superimposition relationship between angle and current, as shown in (1). However, previous studies [22,26] have shown that torque exhibits significant nonlinear behavior. This section and Section 4 analyze the nonlinear characteristics under PM biasing and coil excitation conditions respectively.

$$T = P_l \cdot NI + P_\theta \cdot \theta \quad (1)$$

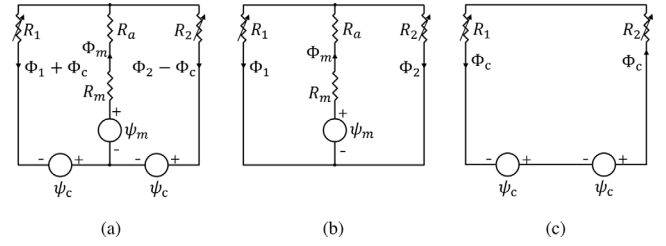


Fig. 3. (a) Magnetic circuit model. It can be separated into (b) PM biased DC magnetic circuit and (c) coil excited AC magnetic circuit.

3.1. Nonlinear modeling

Fig. 3(a) presents the magnetic circuit model of the HRA. Based on the source difference of magnetic flux, it can be separated into DC magnetic circuit (Fig. 3(b)) and AC magnetic circuit (Fig. 3(c)). Herein, R_a , R_m , R_1 and R_2 respectively denote the magnetic reluctance of non-working air gap between PM and mover, the internal PM, and the two working air gaps. The working air gap lengths are approximately as $l_1 = t - h\theta$ and $l_2 = t + h\theta$. Due to the much higher permeability of ferromagnetic materials compared to air, their magnetic reluctance is neglected. The permeability of PM is approximately equal to air. Thus, the magnetic resistances of each part can be calculated as

$$R_1 = \frac{t - h\theta}{\mu_0 S_y}, R_2 = \frac{t + h\theta}{\mu_0 S_y}, R_m = \frac{l_m}{\mu_0 S_m}, R_a = \frac{l_a}{\mu_0 S_m} \quad (2)$$

Let Φ_c and Φ_m represent respectively the flux formed by coil and PM, while Φ_1 and Φ_2 represent the PM flux Φ_m passing through the left and right stator yokes. According to Kirchhoff's current law (KCL), $\Phi_m = \Phi_1 + \Phi_2$. The magnetomotive forces (MMF) of PM and coil can be respectively expressed as $\Psi_m = H_c l_m = \frac{B_r l_m}{\mu_0}$ and $\Psi_c = NI$. According to Fig. 3(b), by Hopkinson's law

$$\Phi_m = \frac{\Psi_m}{R_m + R_a + R_1 // R_2} \quad (3)$$

In the linear model, Φ_m is approximated as constant. However, as displacement changes, $R_1 // R_2$ varies accordingly, rendering Φ_m a nonlinear function. Introducing $k = \frac{S_y}{S_m}$ and substituting into (3)

$$\Phi_m = \frac{2t B_r l_m S_y}{2kt(l_m + l_a) + t^2 - h^2\theta^2} = \frac{M_1}{M_2 - (h\theta)^2} \quad (4)$$

Where $M_1 = 2t B_r l_m S_y$, $M_2 = 2kt(l_m + l_a) + t^2$.

By Kirchhoff's Voltage Law (KVL), $\Phi_1 R_1 = \Phi_2 R_2$, leading to $\Phi_1 = \frac{t+h\theta}{t-h\theta} \Phi_2$. According to $\Phi_m = \Phi_1 + \Phi_2$, Φ_1 and Φ_2 are given by

$$\begin{cases} \Phi_1 = \frac{(t+h\theta) B_r l_m S_y}{2kt(l_m + l_a) + t^2 - h^2\theta^2} = \frac{t+h\theta}{2t} \left(\frac{M_1}{M_2 - (h\theta)^2} \right) \\ \Phi_2 = \frac{(t-h\theta) B_r l_m S_y}{2kt(l_m + l_a) + t^2 - h^2\theta^2} = \frac{t-h\theta}{2t} \left(\frac{M_1}{M_2 - (h\theta)^2} \right) \end{cases} \quad (5)$$

According to the Maxwell stress tensor, the torque is

$$T = h \cdot \frac{\Phi_1^2 - \Phi_2^2}{2\mu_0 S_y} \quad (6)$$

As shown in (5), the PM flux inherently exhibits nonlinear characteristics. The torque expression in (6) further amplifies the nonlinearity, thereby causing significant distortion in the output torque at large deflection angles.

3.2. Finite element analysis of magnetic flux leakage

Considering magnetic flux leakage, the torque is determined by effective flux components of Φ_1 and Φ_2 , which correspond to the perpendicular components of Φ_m at the mover's bottom surface. For small rotations, these are approximately equal to the vertical fluxes

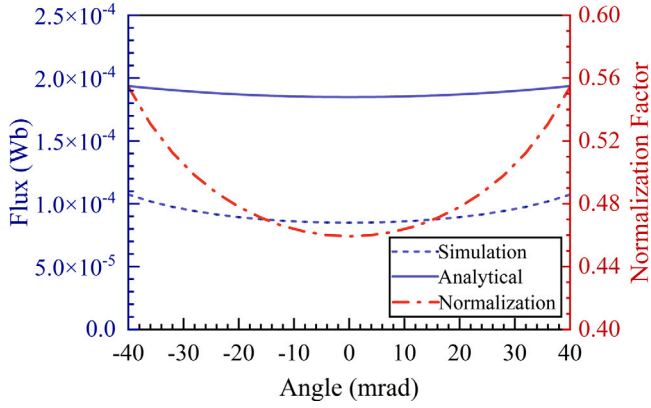


Fig. 4. Analytical value (solid line) and simulation value (dashed line) of the DC flux are compared. The dash-dot line represents the normalization factor.

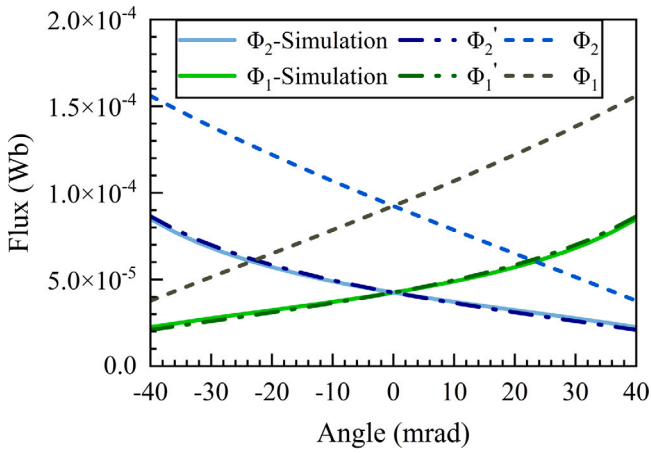


Fig. 5. Comparison between different results of Φ_1 and Φ_2 .

Φ_1' , Φ_2' at stator yokes, simplifying calculation and analysis. Due to flux leakage, only part of Φ_m (denoted as Φ_m') contributes to Φ_1' and Φ_2' , so the sum of Φ_1' and Φ_2' is extracted as Φ_m' in simulation.

The comparison between simulation data (Φ_m') and analytical values of Φ_m is shown in Fig. 4. Combining Table 1, the theoretical calculation expression can be obtained as

$$\Phi_m = \Phi_1 + \Phi_2 = \frac{1.3644 \times 10^{-9}}{7.3815 \times 10^{-6} - (h\theta)^2} (Wb) \quad (7)$$

It is observed that Φ_m' exhibits the same trend as Φ_m . To quantify Φ_m' , a fitting function $y = \frac{a}{b-x^2}$ is employed, obtaining the coefficient of determination $R^2 = 0.9995$. The fitting expression is

$$\Phi_m' = \Phi_1' + \Phi_2' = \frac{1.3685 \times 10^{-10}}{1.6147 \times 10^{-6} - (h\theta)^2} (Wb) \quad (8)$$

Comparing (7) and (8), it can be known that the traditional single leakage coefficient [32] is no longer applicable to the nonlinear model. Therefore, this study introduces a nonlinear model with dual PM leakage coefficients

$$\begin{cases} \Phi_m' = \frac{\alpha \cdot M_1}{\beta \cdot M_2 - (h\theta)^2} \\ \Phi_1' = \frac{t+h\theta}{2t} \left(\frac{\alpha \cdot M_1}{\beta \cdot M_2 - (h\theta)^2} \right) \\ \Phi_2' = \frac{t-h\theta}{2t} \left(\frac{\alpha \cdot M_1}{\beta \cdot M_2 - (h\theta)^2} \right) \end{cases} \quad (9)$$

α and β are PM flux leakage coefficients. For the result of (7) and (8), $\alpha = 0.1003$, $\beta = 0.2187$. The torque expression of the actuator under

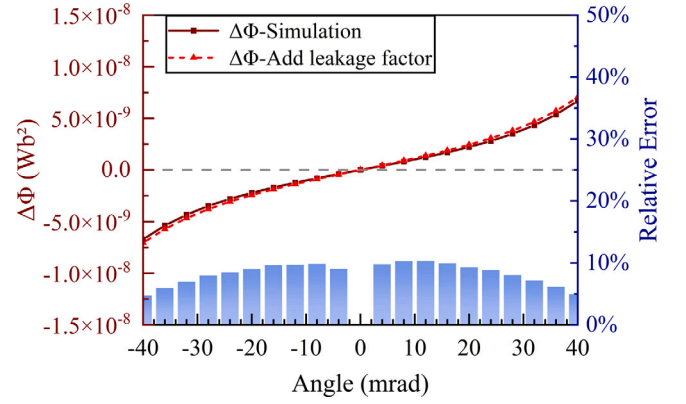


Fig. 6. Comparison between simulation and calculation data of $\Delta\Phi$. The bar chart shows the relative error.

PM biasing condition is

$$T = \frac{h}{2\mu_0 S_y} (\Phi_1' + \Phi_2') (\Phi_1' - \Phi_2') \quad (10)$$

Focus on magnetic flux items in (10). When the leakage coefficient is included, analytical and simulated values for Φ_1 and Φ_2 agree closely (Fig. 5). By setting $\Delta\Phi = (\Phi_1' + \Phi_2')(\Phi_1' - \Phi_2')$, the nonlinear model considering leakage flux achieved a maximum error of 10.3% under deflection of ± 40 mrad (Fig. 6), validating the accuracy of the model.

4. Nonlinear analysis under coil excitation condition

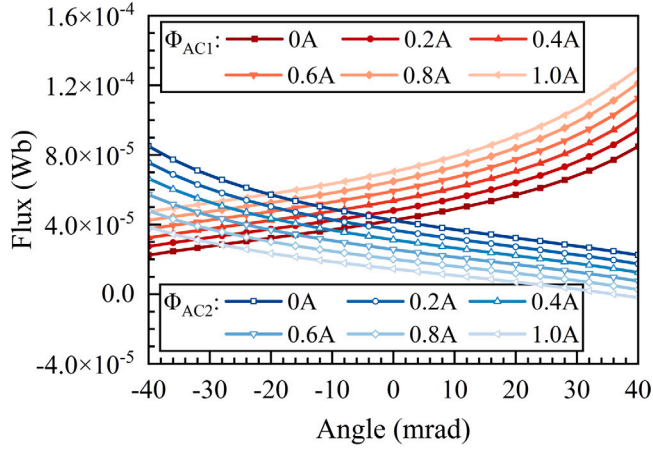
This section further analyzes the coil flux's nonlinear characteristics, constructs the corresponding relationships among nonlinear common-mode and differential-mode components, and thereby develops a high-precision torque nonlinear model. Additionally, the influence of the magnetic saturation on torque is discussed.

4.1. Modification of the nonlinear model

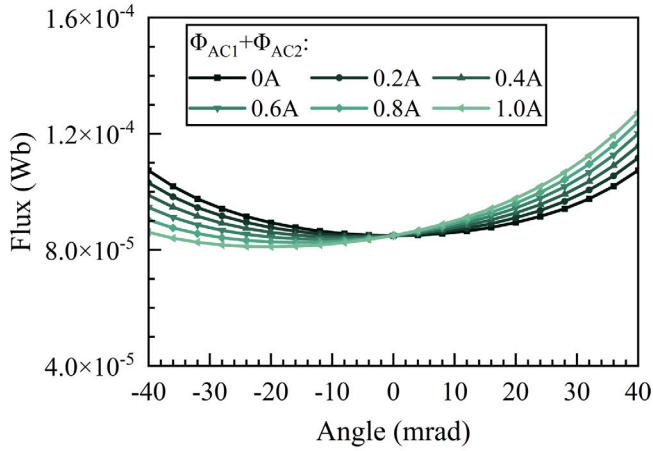
Finite element analysis is carried out under different excitation current conditions. Based on the results, the vertical components of magnetic flux at left and right stator end faces, denoted as Φ_{AC1} and Φ_{AC2} , are extracted (Fig. 7(a)). According to Fig. 3(a), $\Phi_{AC1} = \Phi_1 + \Phi_c$, $\Phi_{AC2} = \Phi_2 - \Phi_c$, thus $\Phi_{AC1} + \Phi_{AC2} = \Phi_1 + \Phi_2$, suggesting this sum should be independent of the current. However, Fig. 7(b) revealed that $\Phi_{AC1} + \Phi_{AC2}$ exhibited a clear current dependence. Therefore, it is necessary to analyze the coil's magnetic flux distribution characteristics in isolation.

The model in Fig. 3(a) is revised to illustrate that the coil generates Φ_{C1} and Φ_{C2} respectively at the two stator end faces, as shown in Fig. 8(a). The corresponding simulation results for Φ_{C1} and Φ_{C2} are shown in Fig. 8(b). Compared with distribution curves of Φ_1 and Φ_2 shown in Fig. 4 and Fig. 5, the results indicate that the two sets of parameters exhibit highly consistent variation patterns. The simulation results for $\Phi_{C1} + \Phi_{C2}$ under coil excitation conditions of 0.2 A, 0.4 A and 0.6 A are fitted to $y = \frac{a}{b-x^2}$, with R^2 are all higher than 0.999, as results shown in (11). Similarly, the simulation results of $\Phi_{C1} - \Phi_{C2}$ are highly consistent with $y = \frac{cx}{b-x^2}$, thereby verifying that Φ_{C1} and Φ_{C2} have the same nonlinear characteristics as Φ_1 and Φ_2 .

$$\begin{cases} \Phi_m' = \Phi_1' + \Phi_2' = \frac{1.3685 \times 10^{-10}}{1.6147 \times 10^{-6} - (h\theta)^2} (Wb) \\ \Phi_{C1} + \Phi_{C2}|_{0.2A} = \frac{1.7596 \times 10^{-11}}{1.5628 \times 10^{-6} - (h\theta)^2} (Wb) \\ \Phi_{C1} + \Phi_{C2}|_{0.4A} = \frac{3.5249 \times 10^{-11}}{1.5735 \times 10^{-6} - (h\theta)^2} (Wb) \\ \Phi_{C1} + \Phi_{C2}|_{0.6A} = \frac{5.3020 \times 10^{-11}}{1.5795 \times 10^{-6} - (h\theta)^2} (Wb) \end{cases} \quad (11)$$



(a)



(b)

Fig. 7. (a) The simulation results of Φ_{AC1} and Φ_{AC2} . (b) The sum of the two exhibits current correlation.

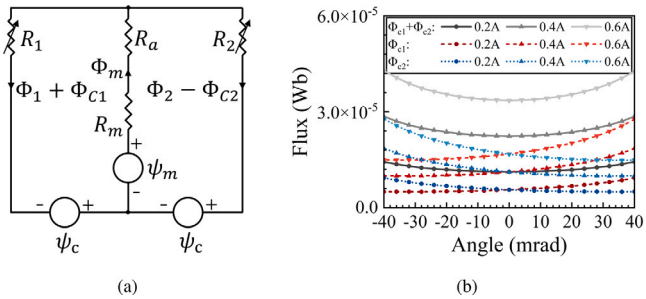
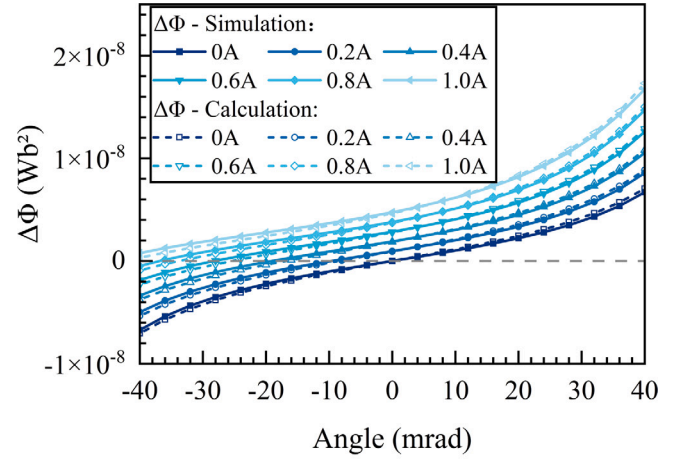
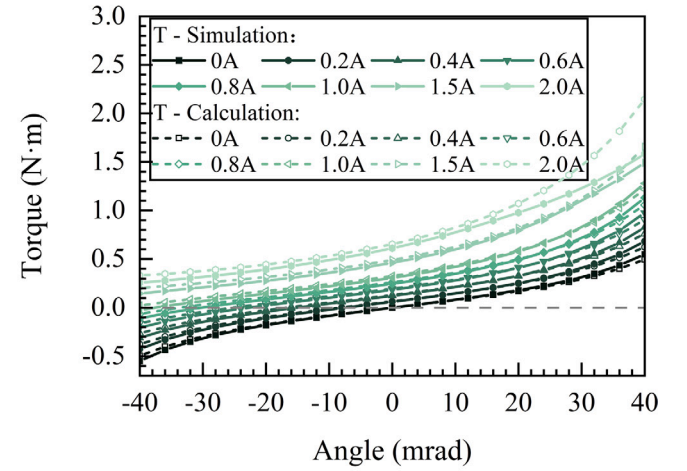


Fig. 8. (a) Schematic diagram of the modified AC magnetic circuit. (b) Simulation results of the coil flux, which is consistent with the variation trend of PM flux.

Defining the common mode and differential mode of PM flux as $\Phi_{PM-C} = \Phi_1' + \Phi_2'$, $\Phi_{PM-D} = \Phi_1' - \Phi_2'$, and the common mode and differential mode of coil flux are $\Phi_{Coil-C} = \Phi_{C1} + \Phi_{C2}$, $\Phi_{Coil-D} = \Phi_{C1} - \Phi_{C2}$. From (11), it is evident that Φ_{Coil-C} and Φ_{PM-C} share nearly the same denominator, indicating a proportion relationship. The proportionality factor is linearly related to current and number of coil



(a)



(b)

Fig. 9. (a) Comparison between simulation and calculation results of flux-related term $\Delta\Phi$ and (b) output torque T .

turns, that is

$$\begin{cases} \Phi_{Coil-C} = \gamma NI \cdot \Phi_{PM-C} \\ \Phi_{PM-D} = \gamma NI \cdot \Phi_{Coil-D} \end{cases} \quad (12)$$

The coil flux leakage coefficient $\gamma = 0.0028$ from (11). Different from the traditional definition of coil flux leakage coefficient, in this study, $\gamma = \frac{1}{NI} \cdot \frac{\Phi_{Coil-C}}{\Phi_{PM-C}}$ is characterized as the proportional relationship between the common mode of coil flux and PM flux.

4.2. Modification of the torque model

According to Fig. 8(a), the torque expression is revised as

$$\begin{aligned} T &= h \cdot \frac{(\Phi_1' + \Phi_{C1})^2 - (\Phi_2' - \Phi_{C2})^2}{2\mu_0 S_y} \\ &= \frac{h}{2\mu_0 S_y} (\Phi_{PM-C} + \Phi_{Coil-D})(\Phi_{PM-D} + \Phi_{Coil-C}) \end{aligned} \quad (13)$$

Focusing on $\Delta\Phi = (\Phi_{PM-C} + \Phi_{Coil-D})(\Phi_{PM-D} + \Phi_{Coil-C})$, Fig. 9(a) compares the calculation and simulation results. It is indicated that under 0–1 A current condition, the error is less than 10%.

The corrected magnetic flux model reveals that all common-mode components have the same nonlinear characteristic, expressed as $y = \frac{a}{b-x^2}$, while all differential-mode components follow the form $y = \frac{cx}{b-x^2}$.

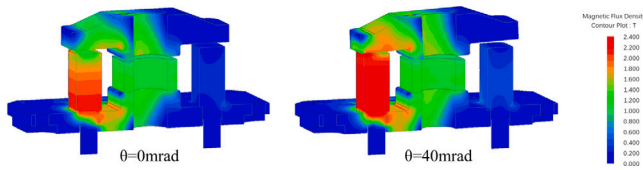


Fig. 10. The cross-sectional profile of magnetic flux density inside actuator (coils are not shown) under 2 A excitation.

The interaction of these components gives rise to the unique nonlinear characteristics of HRA. Based on the above equations, this study derives the following expression for the nonlinear torque model of HRA

$$T = \frac{h(1 + \gamma N I \frac{h\theta}{2t})(\gamma N I + \frac{h\theta}{t})}{2\mu_0 S_y} \left(\frac{\alpha M_1}{\beta M_2 - (h\theta)^2} \right)^2 \quad (14)$$

$$\text{Where } M_1 = 2tB_r l_m S_y, M_2 = 2kt(l_m + l_a) + t^2.$$

4.3. Torque nonlinearity caused by magnetic saturation

Fig. 9(b) compares theoretical and simulated torque outputs. The model (14) exhibits high accuracy (error < 0.07 N m) within 1 A current range. However, errors increase for 1.5–2 A, especially at large angles. The reason is that magnetic saturation causes permeability reduction in soft magnetic materials, thereby constraining flux enhancement (Fig. 10). Compared with the middle position, when $\theta = 40$ mrad, the left stator yoke reaches the saturation state. Hence, to guarantee the effective application of nonlinear model, the effective current range is limited to 0–1 A.

5. Mechanical design

In the HRA-FSM system, the mechanical design is crucial for achieving high overall system performance. However, it faces two major challenges: First, the HRA exhibits pronounced nonlinear torque and negative stiffness characteristics. According to Fig. 9(b), the maximum negative stiffness in the 0 A state reaches 13.74 N m/rad. Second, the small air gap design brings a significant attractive force, reaching 71.4 N in the 0 A balanced state. To avoid potential wear of the pivot bearing structure and mirror surface deformation of the mirror-compliant mechanism bonding form [23], the mechanical structure proposed in this paper adopts a combined form of compliant mechanism and high-stiffness mirror support structure.

The schematic diagram of the compliant mechanism is shown in Fig. 11, which consists of two parts: Compliant mechanism I (abbreviated as CM-I) and Compliant mechanism II (abbreviated as CM-II). CM-I supplies the majority of the x -axis positive stiffness, though its rod-shaped design inadequately constrains motion along the non-driven axis (y -axis). To ensure high stiffness in y -axis direction, CM-II is incorporated, which also contributes a minor portion of the x -axis positive stiffness. The resulting compliant mechanism provides torsional stiffness of 14.63 N m/rad slightly exceeding the HRA's maximum negative stiffness, thereby preventing the snap-in of the mover to the yoke when operated in open-loop.

CM-I is bonded to the mover through an adapter, while CM-II is rigidly connected to the mover and mirror support structure via screws. The mirror has screw holes at the four corners and is also connected to the mirror surface support structure by screws. The entire compliant mechanism is installed within an aluminum enclosure with sufficient strength. To ensure elastic deformation within the ± 41.5 mrad deflection range, the compliant mechanism is fabricated from Ti-6Al-4V alloy, which exhibits a high yield strength of 827 MPa. The FEA conducted using ANSYS Workbench simulated the stress distribution under a deflection of 41.5 mrad. As shown in Fig. 12, the maximum stress occurs at CM-II with a peak value of 430.4 MPa, which is well below the yield strength of the Ti-6Al-4V.

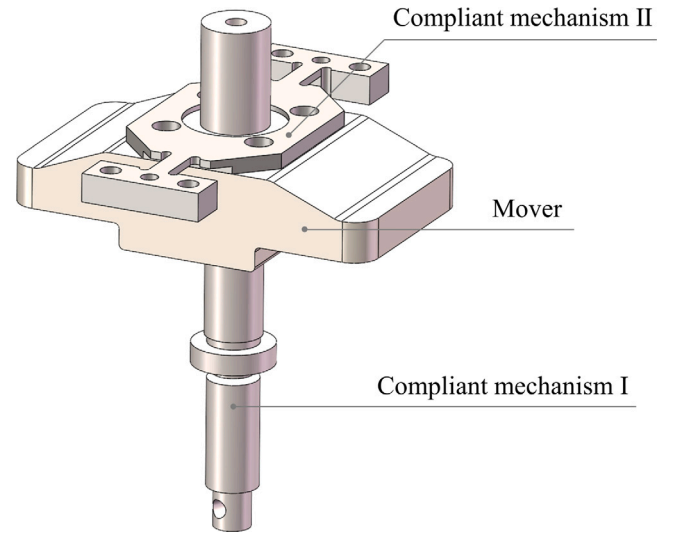


Fig. 11. Schematic diagram of the compliant mechanism.

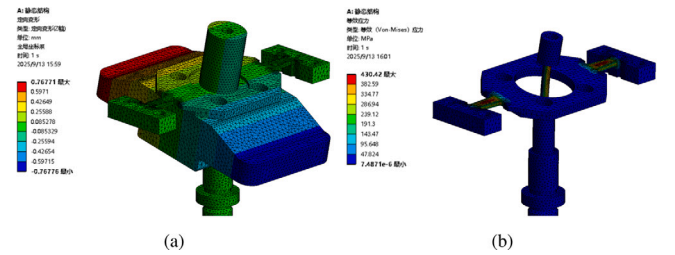


Fig. 12. Finite element simulation results of displacement in z -axis direction (a) and stress (b) for the compliant mechanism under 41.5 mrad deflection.

6. Magnetic flux evaluation experiment

The nonlinear torque characteristics fundamentally originate from magnetic flux nonlinearity. Given the challenges in direct torque measurement, Fig. 9 demonstrates that the error trends of $\Delta\Phi$ and T between simulated and theoretical torque models exhibit strong consistency. This indicates that torque nonlinearity can be effectively evaluated by magnetic flux measurement.

6.1. Evaluation principle

As a point measurement approach, Hall sensors capture local magnetic flux density (B), requiring conversion to flux ($\Phi = B \cdot S$) for analysis. Distinct from translational HRAs, the deflection motion in this actuator results in significant spatial variations in the magnetic flux density distribution on the end faces of yokes.

$$\delta_i(\theta) = \frac{\Phi_i(\theta)}{\Phi(\theta)} \quad (15)$$

To address the challenge of spatial variation, FEA was employed to evaluate the magnetic flux density distribution at measurement points along the stator end faces (the points are also located on yellow line in Fig. 14(a)). For each measurement point, the distribution curve $B_i(\theta)$ was obtained, and the corresponding flux was calculated as $\Phi_i(\theta) = B_i(\theta) \cdot S_y$. Normalization using (15) enabled identification of the measurement region with minimal fluctuation error in the ratio, where $\Phi(\theta)$ denotes the vertical component of magnetic flux density on stator end faces. As shown in Fig. 13, the normalization factor variation remains below 2% within a distance range of 1 to 2 mm from the stator yokes' inner surfaces, meaning that $\delta_i(\theta) \approx \text{constant}$ ($\delta = 0.855$).

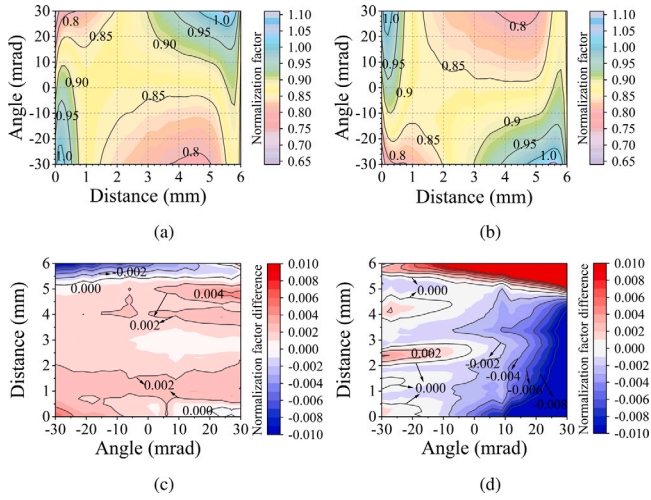


Fig. 13. Normalization factor $\delta_i(\theta)$ at left (a) and right (b) stator measurement points under 0 A. Their variations (c) (d) after 1 A energization demonstrating negligible influence of current on flux distribution.

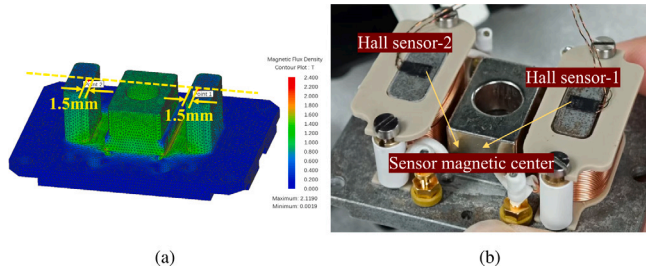


Fig. 14. (a) The meshing of yoke end face is refined, and ideal measurement locations are ascertained via simulation. (b) Installation layout of the Hall sensors based on results.

Based on these findings, the optimal center measurement point for Hall sensors was determined to be 1.5 mm, as illustrated in Fig. 14(a).

6.2. Experimental validation

The small working air gap of actuator and large-span variation characteristics of magnetic flux density impose stringent requirements on Hall sensors. In this study, the HE144T linear Hall sensors from Asensor Technology were employed, featuring a thickness of only 0.5 mm and a measurement range exceeding ± 2 T. The installation location of the linear Hall sensor is depicted in Fig. 14(b). Before installation, the Hall sensors were calibrated using a reference gaussmeter with an accuracy of $\pm 1\%$. Repeated measurements across five cycles resulted in a standard deviation below 1% of the measured value, confirming the robustness of the experimental validation. Alignment was verified under a microscope to within 0.1 mm. To avoid potential damage during mirror movement, the Hall sensors were removed in control experiment.

A comparison of measurement and simulation results is shown in Fig. 15. The Hall sensor measurements (Fig. 15(a)) show strong agreement with simulations, revealing slight flux nonlinearity versus deflection angle. The superimposed results of Hall sensor-1 and Hall sensor-2 (Fig. 15(b)) validate the nonlinear modeling theory, where the combined signal $\Phi_{PM-C} + \Phi_{Coil-D}$ exhibits current-dependent deflection. Additionally, the full stroke cycle measurement captures hysteresis-induced non-coincidence characteristics. The measurement result error ranges from 15% to 17% ($\delta = 0.83$ – 0.85), it is highly consistent with

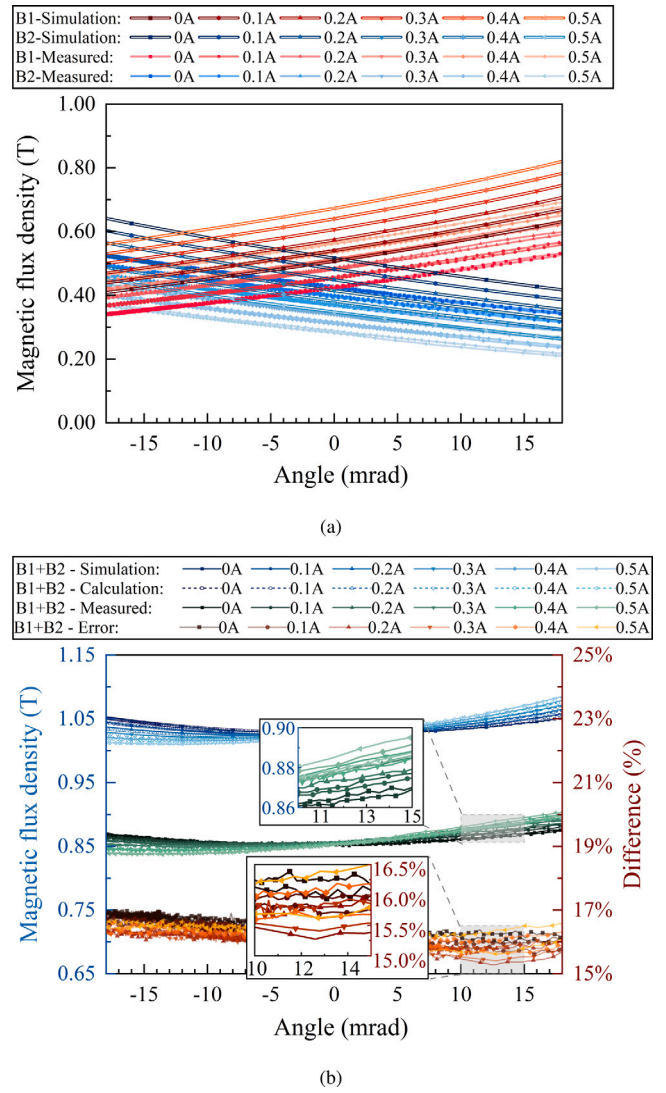


Fig. 15. (a) Dual Hall sensors measurements vs simulation (0–0.5 A)(b) Comparison of the superimposed measurement data with the simulation.

the simulation result $\delta = 0.855$ within 3% maximum error. Furthermore, although the hysteresis effect introduces additional measurement errors, the fluctuation error is less than 2%, thereby the method successfully validates the Hall sensor-based approach for characterizing HRA deflection nonlinearities.

7. Control experiment

In contrast to inverse control approaches that depend on state observers for uncertainty estimation, this work implements a forward control strategy based on the nonlinear model. The following section presents experimental comparison results obtained before and after linearization.

7.1. Experimental setup

Fig. 16 illustrates the experimental platform and block diagram. The FSM features a rotational inertia of $3 \times 10^{-5} \text{ Kg m}^2$ with a ± 41.5 mrad deflection range. The compliant mechanism is fabricated from Ti-6Al-4V alloy, selected for its high fatigue strength to ensure long-term reliability under high-cycle loading. A linear current amplifier (OPA549) is employed to drive the actuator. The resistance of actuator is 3.2Ω .

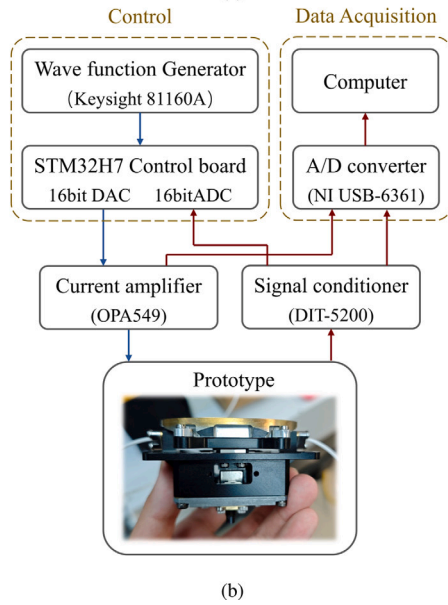
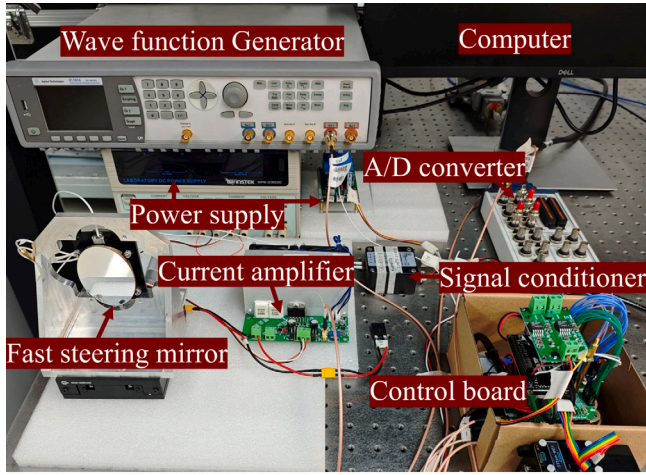


Fig. 16. (a) Hardware layout of HRA-FSM experimental platform. (b) Block diagram of the platform.

To ensure safe heat dissipation for the current amplifier, a $5\ \Omega$ power resistor is connected in series with actuator. In the control experiment, the Hall sensors are removed, and displacement sensing is achieved solely by eddy current sensors. The eddy current sensors employed a novel differential layout, with the measurement target being a high-stiffness mirror support structure. The voltage-displacement sensitivity is calibrated by optical autocollimator. This measurement approach has been compared and validated with the grating shear interferometry method [34]. The controller is designed based on STM32H7 MCU with a sampling frequency of 20 KHz (the corresponding sampling time is 0.05 ms). Peripheral circuits utilize AD7606 and AD5676 for analog-to-digital and digital-to-analog conversion. Control signals are supplied by a function generator (Keysight 81160 A). The computer acquires the input and output signals via the NI USB-6361 data acquisition module for data processing.

7.2. Open-loop test

A sinusoidal sweep signal with an amplitude of 50 mV and a frequency ranging from 1 Hz to 801 Hz is generated by the function generator and applied to the current amplifier (conversion rate: 1 A/V).

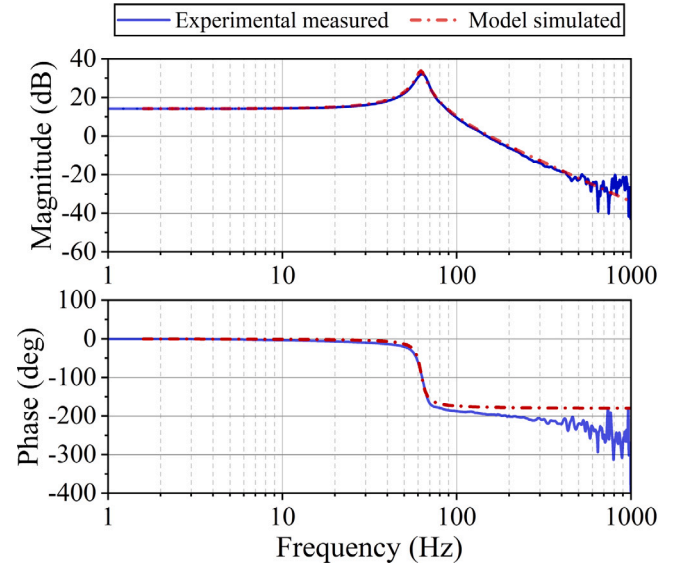


Fig. 17. Schematic diagram of the system's open-loop frequency response.

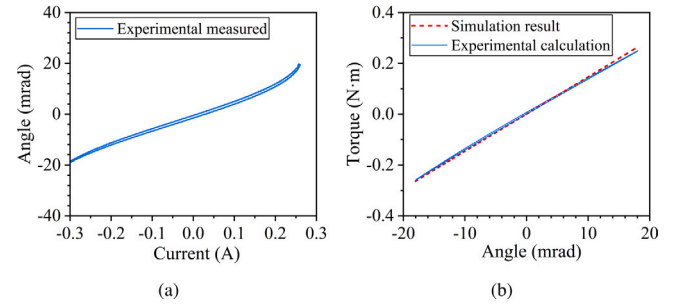


Fig. 18. (a) Quasi-static open-loop current-angular displacement curve. (b) Comparison between calculation results of the torque nonlinear model and simulation results of the compliant mechanism.

The input signal and displacement sensor signal are synchronously collected by data acquisition module, and the open-loop frequency response characteristics of the system are identified through Fourier transform. The results are shown in Fig. 17, it can be seen that the frequency response of the system is second-order and the resonant frequency is 64 Hz. Through the identification of the system, the resulting system transfer function is $G(s) = \frac{7.916 \times 10^5}{s^2 + 40.85s + 1.539 \times 10^5}$. In addition, the phase-frequency curve shows a high-frequency phase lag caused by the eddy current effect and dynamic characteristic of the current amplifier [35].

Under open-loop condition, a 0.2 Hz quasi-static excitation signal is applied to the HRA-FSM. To avoid the snap-in of the mover to the yoke when operated in open-loop, the stroke is regulated to be within ± 18 mrad. At this time, the driving torque is balanced with the restoring torque of the compliant mechanism. Fig. 18(a) shows the relationship between the rotation angle of HRA-FSM and the corresponding current. It can be seen that the two present a significant nonlinear relationship, corresponding to the nonlinearity of the actuator output torque. The reason for the asymmetry of the curve is slight processing and installation errors. In addition, the figure shows that the system has a hysteresis effect, with the maximum width of the hysteresis loop being approximately 3.1% of the motion stroke.

Fig. 18(a) can verify the simulation results of the compliant mechanism because driving torque is balanced with the restoring torque of the compliant mechanism. Substitute the result in Fig. 18(a) into (14) for calculation and compare with the simulation value, the result shown in

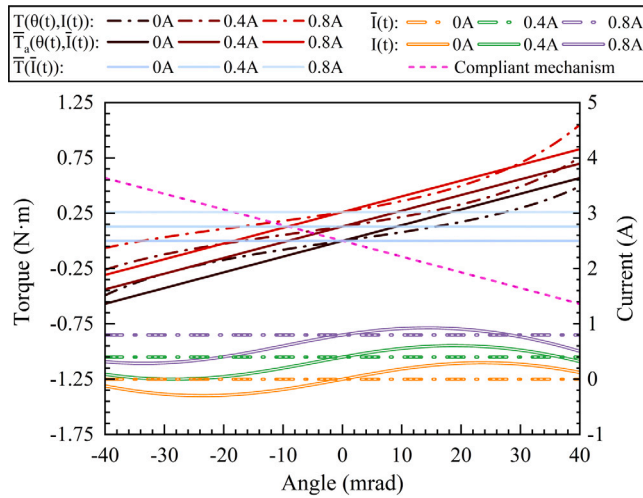


Fig. 19. A comparison of actuator output torque versus input current with and without compensation.

Fig. 18(b). It can be seen that the calculation result is 14.09 N m/rad while the simulation value is 14.63 N m/rad. Therefore the 3.7% error verifies the accuracy of the nonlinear model.

7.3. Controller design and experimental validation

$$\begin{cases} T(\theta(t), I(t)) = \frac{h(1+\gamma N I \frac{h\theta}{2t})(\gamma N I + \frac{h\theta}{2t})}{2\mu_0 S_y} \left(\frac{\alpha M_1}{\beta M_2 - (h\theta)^2} \right)^2 \\ T(\theta(t), I(t)) = J \frac{d^2\theta(t)}{dt^2} + c \frac{d\theta(t)}{dt} + k\theta(t) \end{cases} \quad (16)$$

The system's torque balance Eq. (16) is established, wherein J , c , k_m denotes the moment of inertia, structural damping coefficient, and torsional stiffness of the compliant mechanism, respectively. It shows that the HRA-FSM is a nonlinear system, and direct application of conventional linear control algorithms results in degraded performance. To address this, a compensator is designed to linearize the output torque, thereby transforming the system into a second-order linear system. Specifically, set the torque expression of the HRA after feedback linearization as

$$\bar{T}_a(\theta(t), \bar{I}(t)) = P_I \cdot \bar{I}(t) + P_\theta \cdot \theta(t) \quad (17)$$

According to (14), when the mover is in balanced position, $T_{\theta=0} = \frac{\alpha^2 M_1^2 \gamma N h}{2\beta^2 M_2^2 \mu_0 S_y} \cdot I$, in order to effectively simplify the system model, setting $P_I = \frac{\alpha^2 M_1^2 \gamma N h}{2\beta^2 M_2^2 \mu_0 S_y}$, $P_\theta = k_m$ (in this paper, $P_I = 0.308$ N m/A, $P_\theta = 14.63$ N m/rad), then, after feedback linearization, the system torque model $T(\bar{I}(t))$ can be simplified as $T(\bar{I}(t)) = P_I \cdot \bar{I}(t)$. This approach reduces the difficulty of controller design and parameter adjustment. Here, $\bar{I}(t)$ is the input signal of this linear system, and $\bar{I}(t)$ can be obtained by constructing the mapping relationship between the input signals $I(t)$ and $\bar{I}(t)$ of the original nonlinear system. Let (14) equal (17), get

$$\begin{cases} I(t) = \frac{-b \pm \sqrt{b^2 - 4ac + \frac{4a}{d}(P_I \cdot \bar{I}(t) + P_\theta \cdot \theta(t))}}{2a} \\ a = \frac{\gamma^2 N^2 h^2 \theta}{4\mu_0 S_y t} \\ b = \frac{\gamma N h}{2\mu_0 S_y} + \frac{\gamma N h^3 \theta^2}{4\mu_0 S_y t^2} \\ c = \frac{h^2 \theta}{2\mu_0 S_y t} \\ d = \left(\frac{\alpha \cdot 2t B_s I_m S_y}{\beta \cdot (2kt(I_m + I_a) + t^2) - (h\theta)^2} \right)^2 \end{cases} \quad (18)$$

The angle value $\theta(t)$ can be obtained by real-time acquisition of the sampling data from the signal conditioner. Since the value of $\theta(t)$ is

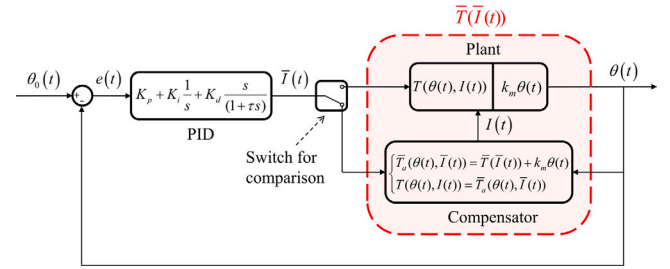


Fig. 20. The control block diagram. The switch enables a performance comparison with and without compensation. The red box representing the compensated linear system.

relatively small, throughout the entire angular travel interval, there is $b^2 - 4ac + \frac{4a}{d}(P_I \cdot \bar{I}(t) + P_\theta \cdot \theta(t)) > 0$, meaning that (18) always has roots. The selection of roots (determination of '±') can be verified by substituting them into (14). Fig. 19 shows the simulation results of nonlinear HRA torque model $T(\theta(t), I(t))$, linearized HRA torque model $\bar{T}_a(\theta(t), \bar{I}(t))$, linear system model $T(\bar{I}(t))$ and the input signals $I(t)$ and $\bar{I}(t)$.

The control block diagram is shown in Fig. 20, where the switch enables a comparison of system performance with and without compensation. A well-established proportional–integral–derivative (PID) control algorithm is employed for its computational simplicity, ensuring real-time performance on embedded hardware. The controller parameters are obtained through iterative tuning: initially, K_i and K_d are set to 0, while K_p is increased until a slight overshoot (<10%) occurred. Then, raise K_i to eliminate steady-state error until the system settles promptly without significant oscillation. With K_p and K_i fixed, gradually increase K_d to suppress overshoot. To prevent the amplification of high-frequency noise due to excessive K_d , a low-pass filter with a cutoff frequency $f = 2000$ Hz is implemented. K_d could be increased until the steady-state output remained stable without noise-induced jitter, then K_p and K_i can be appropriately increased. The control parameters are ultimately determined as $K_p = 60$, $K_i = 1200$, and $K_d = 0.14$ to ensure stable performance. The program is set to limit the output current to ± 1 A to prevent the HRA from experiencing magnetic saturation. To evaluate the effectiveness of the nonlinear compensator, set the reference motion to a sinusoidal sweep signal with an amplitude of 1 mrad, a duration of 40 s and a frequency ranging from 1 Hz to 801 Hz, and comparative tests are performed on the FSM platform at deflection angles of 10 mrad and 30 mrad (Fig. 21). The experimental results demonstrate that, without compensation, the PID controller exhibits unstable performance across different deflection angles; specifically, the lowest point in amplitude response curve within 100 Hz varies from -1.1 dB to -2.5 dB and bandwidth variation reaches 84 Hz. With the nonlinear compensator implemented, the system achieves significantly stabilized control performance throughout the tested angles; the curves almost coincide in the low-frequency regime, and bandwidth variation achieves 48.8% reduction.

However, the proposed feedback linearization compensation method only addresses the amplitude of the motion, the phase-frequency characteristics of the compensated system are almost unchanged. Furthermore, as the frequency increases, the frequency-dependent eddy current and hysteresis effects become dominant, substantially influencing both the amplitude–frequency and phase-frequency characteristics of the system. Therefore, to effectively evaluate the proposed feedback linearization compensation method, a ± 15 mrad triangular wave and a ± 30 mrad sinusoidal wave with frequency of 10 Hz are selected as references, and the system's tracking performance is tested with and without compensation at 10 mrad deflection. The comparison results are shown in Fig. 22. Affected by the limited output current, the peak tracking error of HRA-FSM is relatively

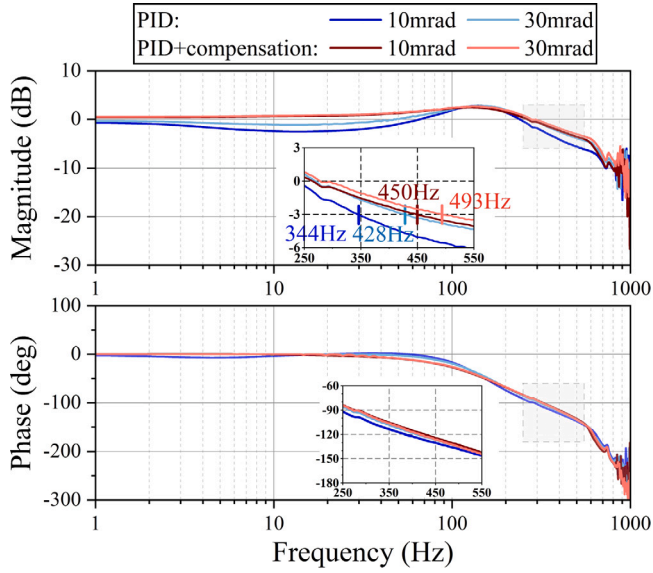


Fig. 21. A comparison diagram of the system's amplitude-frequency and phase-frequency characteristics with and without compensation at different deflection angles.

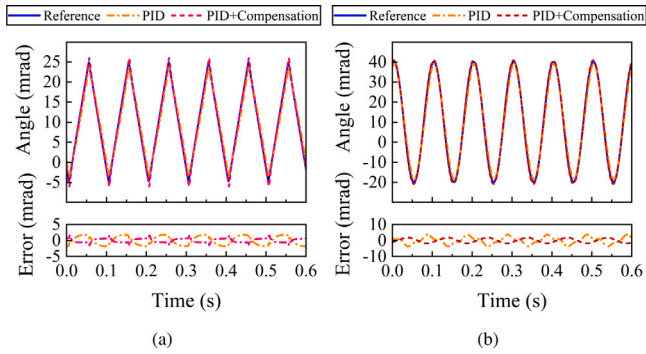


Fig. 22. Comparison of (a) triangular wave and (b) sinusoidal wave tracking results for frequency of 10 Hz.

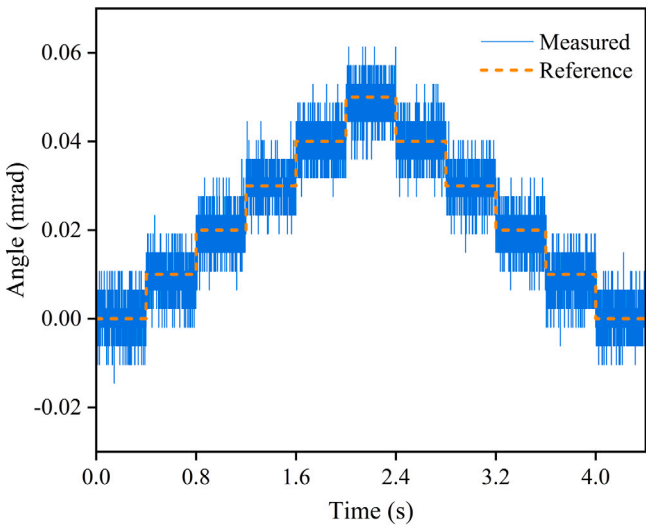


Fig. 23. Small step response with 10 μ rad steps.

Table 2

Performance comparison with typical FSMs.

Ref.	Aperture (mm)	Range (mrad)	Bandwidth (Hz)	RBP (mrad Hz)	FSM type
This work	50.8 $\times$ 71.8	± 41.5	450	18,675	HRA
TT60SM [36]	$\Phi 25$	± 12	<550	<6600	PEA
RC150S [37]	$\Phi 15$	± 2.5	<2200	<5500	PEA
[38] [39]	45 $\times$ 35	± 31.4	375	11,775	VCM
FSM850.01 [40]	$\Phi 50$	± 17.5	180	3150	VCM

significant. To isolate phase delay and enable accurate comparison of tracking errors, the measurement results $\theta_m(t)$ are subjected to a time shift transformation [9]: $\theta_m'(t) = \theta_m(t - t')$, where t' is the time delay between the reference signal $\theta_r(t)$ and the measurement results $\theta_m(t)$. The root mean square (RMS) error is defined as

$$\delta_{rms} = \frac{\sqrt{\frac{1}{T} \sum (\theta_r(t) - \theta_m'(t))^2}}{\max(\theta_r(t)) - \min(\theta_r(t))} \times 100\% \quad (19)$$

According to the results in Fig. 22, the RMS tracking errors of the compensated system are 2.06% for triangular wave and 2.04% for sinusoidal wave, representing 36.4% and 25.1% reductions compared to pre-compensation performance respectively.

Small step command is adopted to test the deflection resolution of the system, the deflection command is set to 10 μ rad steps. The tracking result shown in Fig. 23 indicates that the positioning resolution of the system is at micro-radian level, and the system disturbance error is 4.3 μ rad.

For the same type of single-axis FSM, Table 2 compares performance of the proposed HRA-FSM with typical FSMs, where the bandwidths of PEA-FSMs are specified conservatively below their resonant frequency. Additionally, based on the known data, the load mass and rotational inertia of the proposed FSM are both greater than those of the listed FSMs. The stroke-bandwidth product (RBP), which reflects the ability to simultaneously achieve large deflection and high-speed response, is adopted for comprehensive performance evaluation [3]. As demonstrated, although the heavier load reduced the achievable bandwidth, the proposed FSM achieves the highest RBP, enabling higher dynamic tracking or larger aperture in airborne and astronomical imaging platforms.

8. Conclusion

This study develops a large-aperture HRA-FSM prototype to investigate torque nonlinearity. Theoretical modeling reveals that both PM flux and coil flux contribute to torque nonlinearity, with a concise correspondence between their common-mode and differential-mode components. The established nonlinear torque model achieves 0.07 N·m maximum error compared to simulation results. A linear Hall sensor measurement experiment was then designed to successfully validate the nonlinear model with <3% measurement error. Finally, the implementation of feedback linearization control reduces bandwidth fluctuations by 48.8% and tracking errors up to 36.4%. Through benchmarking against typical FSMs' performance metrics, this study successfully extends the HRA-FSM's technological advantages from 25 mm to 50 mm optical aperture applications.

However, the nonlinear modeling process is based on static magnetic field distributions and its applicability under high-frequency conditions has not been fully studied, future work will focus on dynamic variation characteristics of magnetic flux under high-frequency conditions and integrating compact Hall sensors for real-time feedback.

CRediT authorship contribution statement

Wei Gao: Writing – original draft, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Yongsen Xu:** Validation, Supervision, Resources, Project administration, Methodology, Investigation. **Rui Xu:** Visualization, Project administration, Formal analysis. **Fuchao Wang:** Validation, Investigation. **Yulei Xu:** Investigation, Formal analysis. **Chao Liang:** Visualization, Investigation. **Ping Jia:** Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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